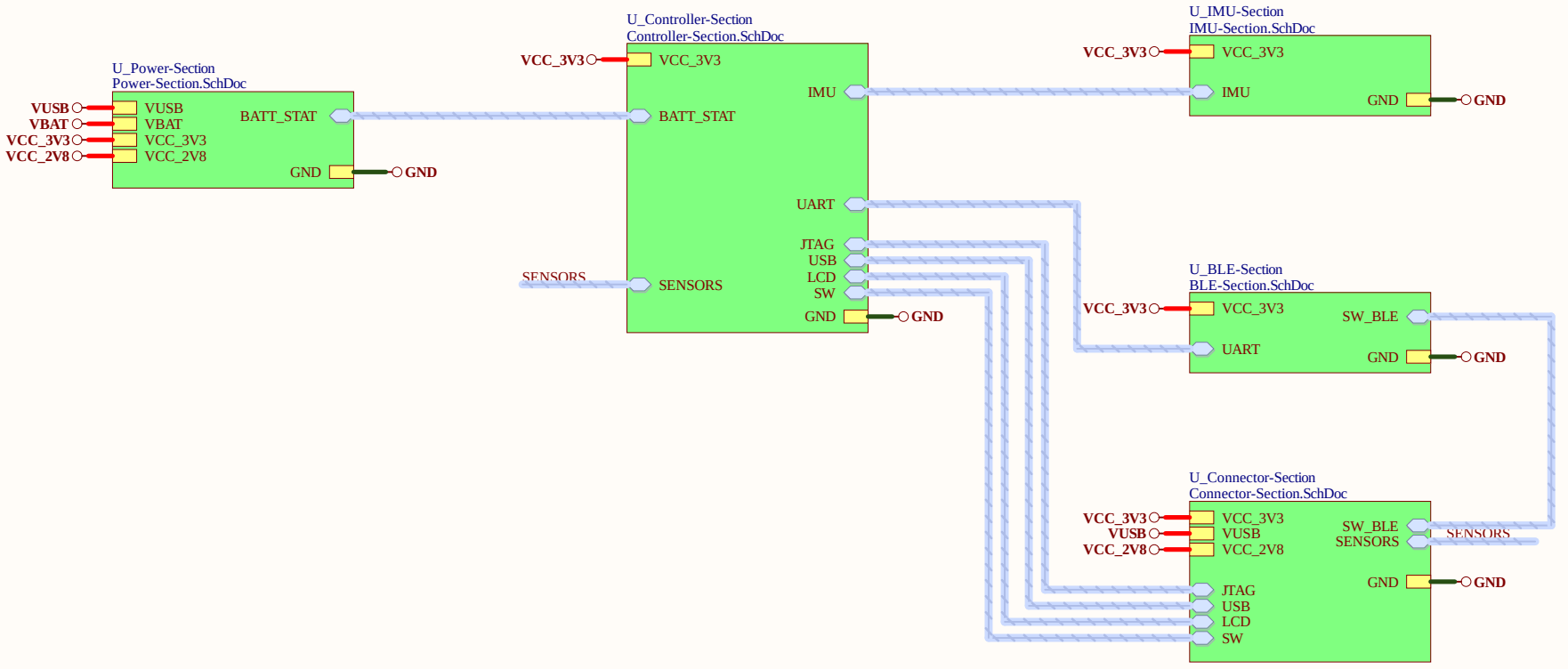
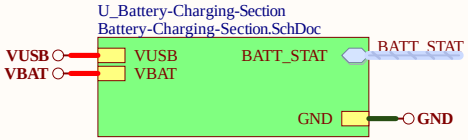
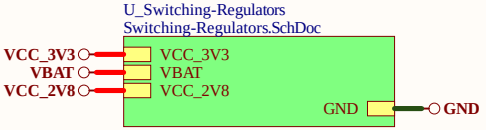


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	Size: A4	Number: 1	Revision: 1
	Date: 6/30/2022	Sheet 1 of 8	
	File: C:\Users\..\Top-Sheet.SchDoc	Drawn By: Huzaifa Ali	









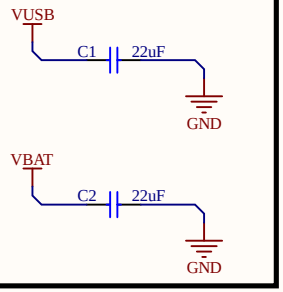
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	Size: A4	Number: 1	Revision: 1
	Date: 6/30/2022	Sheet 2 of 8	
	File: C:\Users\..\Power-Section.SchDoc	Drawn By: Huzaifa Ali	

BATTERY STATUS

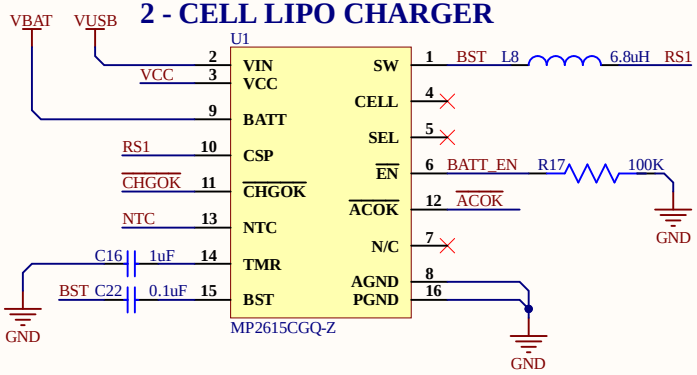


CELL is floating for 2 cells
SEL pin is Floating for 4.2/cell terminating voltage
IC=52.5mV/Rs1 mOhm
IC=2.1A
NTC sensing not used.

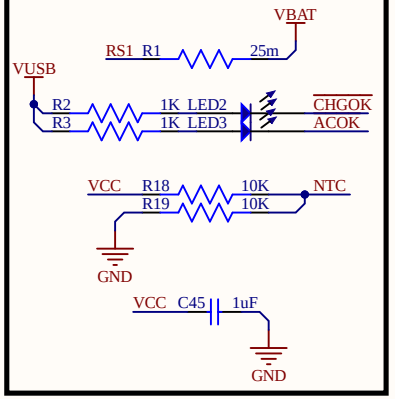
DECOUPLING CAPS

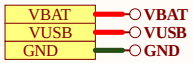


2 - CELL LIPO CHARGER



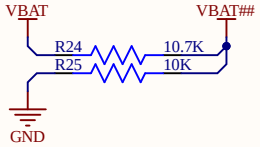
SUPPORTING CIRCUIT



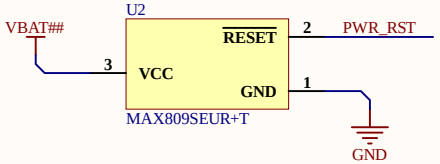


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	Size: A4	Number: 1	Revision: 1
	Date: 6/30/2022	Sheet 3 of 8	
	File: C:\Users\..\Battery-Charging-Section.SchDoc	Drawn By: Huzaifa Ali	

VDR FOR 6V to 2.9V

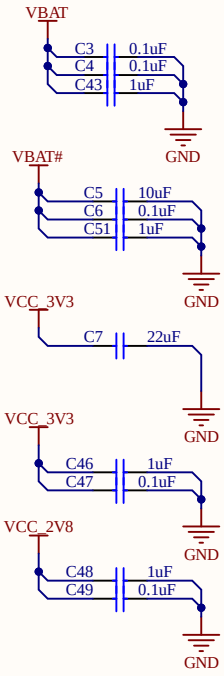


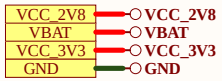
2.9V BATTERY SUPERVISOR



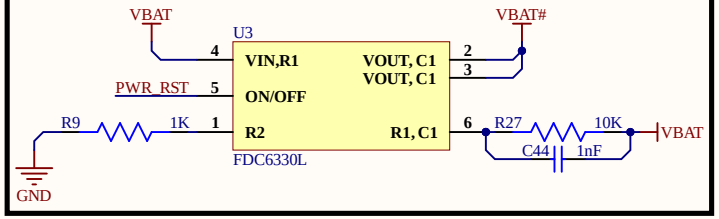
RESET Output remains low while VCC is below the reset threshold, and for at least 140ms after VCC rises above the reset threshold.
Datasheet pg 5

DECOUPLING CAPS



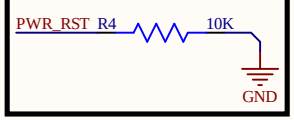


HIGH-SIDE SWITCH

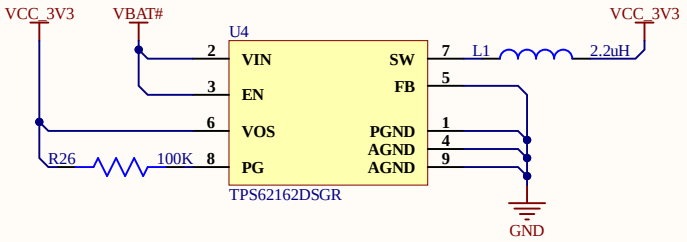


U3 will turn off when ON/OFF is low.
U3 will be enabled when ON/OFF is high.
VON/OFF= 1.5V-8V
VIN= 3-20V

R-PULL DOWN

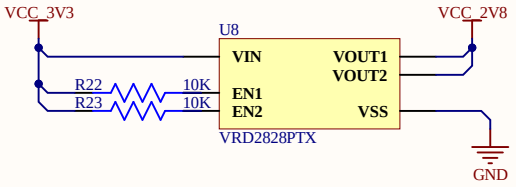


3V3 BUCK REGULATOR



Fixed regulator.
VOUT= 3V3/1A

2V8 REGULATOR



Cannot open file D:\eSpark\eSpark-Hardware\Altium Template\espark-consulting-group-150x150.jpg. File does not exist.

Title	ArmBand-Control-Board.PrjPcb		
Size	Number		Revision
A4	1		1
Date:	6/30/2022	Sheet 4 of	8
File:	C:\Users\..\Switching-Regulators.SchDoc	Drawn By:	Huzaifa Ali

BATTERY STATUS



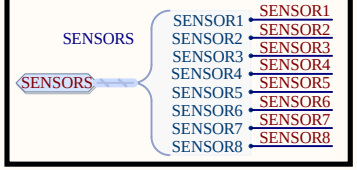
IMU SIGNALS



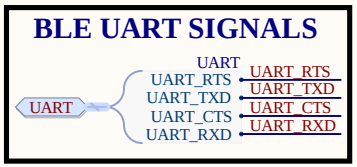
USB SIGNALS



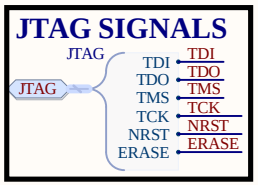
SESNOR SIGNALS



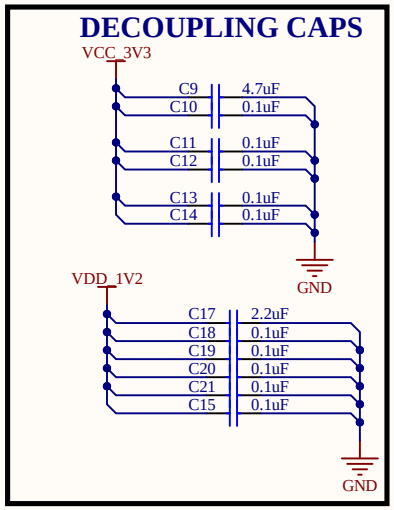
BLE UART SIGNALS



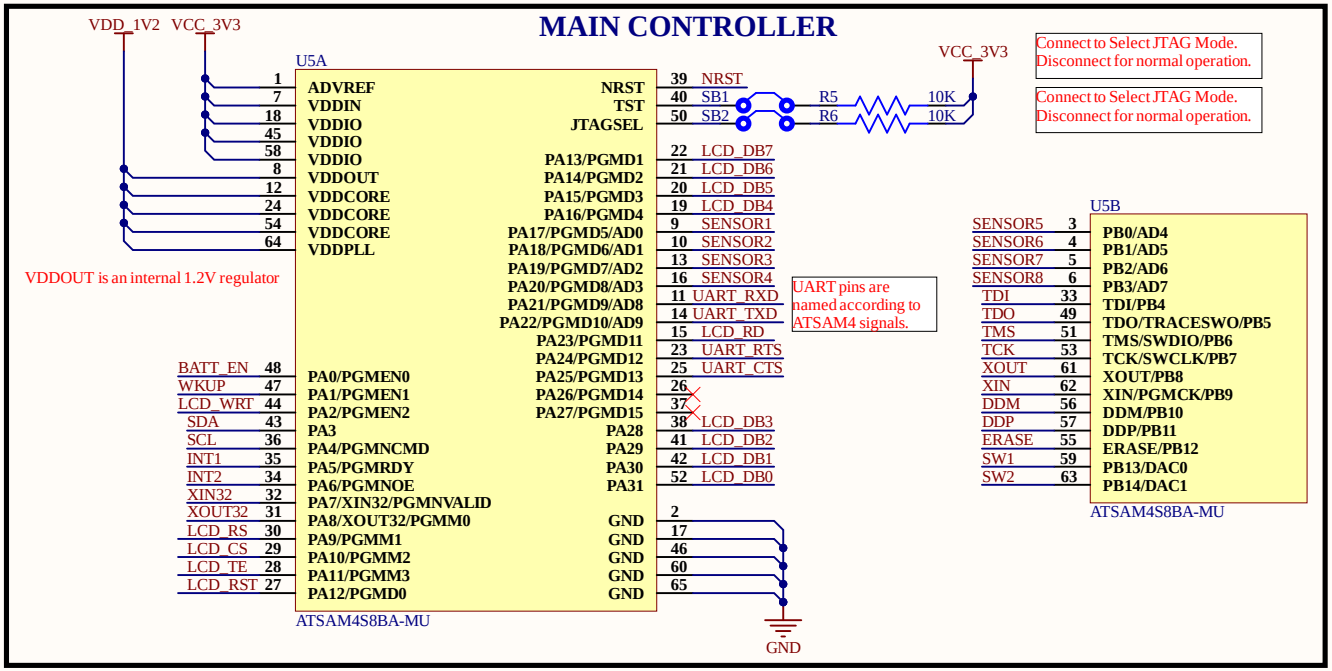
JTAG SIGNALS



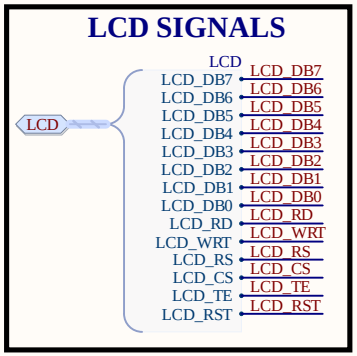
DECOUPLING CAPS



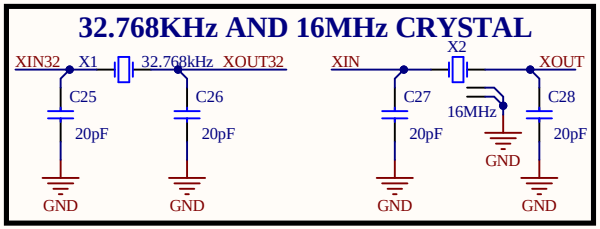
MAIN CONTROLLER



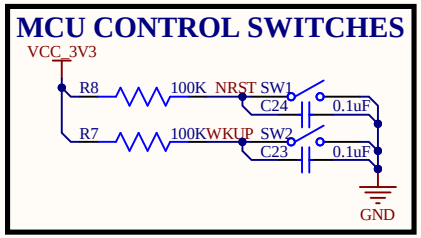
LCD SIGNALS



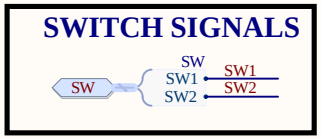
32.768KHz AND 16MHz CRYSTAL



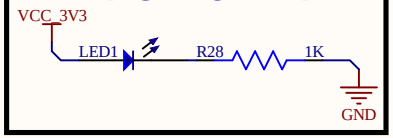
MCU CONTROL SWITCHES

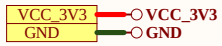


SWITCH SIGNALS



INDICATION LED





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Title	ArmBand-Control-Board.PrjPcb	
Size: A4	Number: 1	Revision: 1
Date: 6/30/2022	Sheet 5 of 8	
File: C:\Users\..\Controller-Section.SchDoc	Drawn By: Huzaifa Ali	

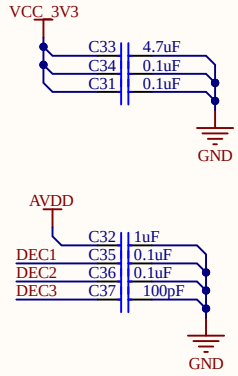
BLE UART SIGNALS



SW SIGNALS

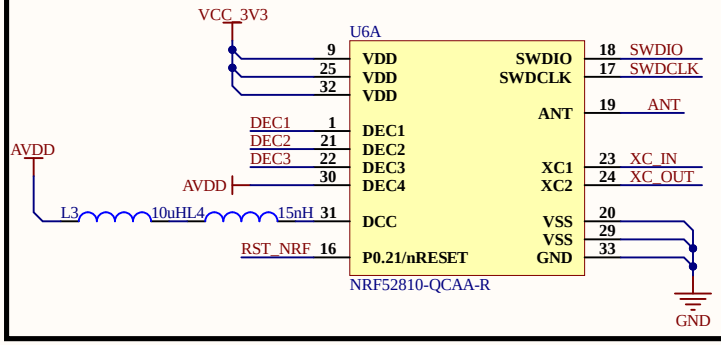


DECOUPLING CAP

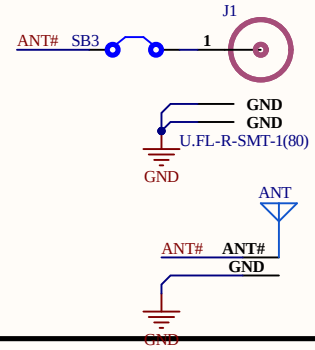


C36 may be disconnected if not needed

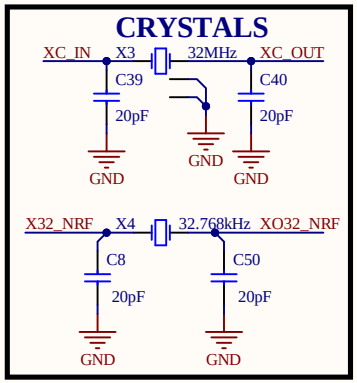
BLE TRANCIEVER POWER BANK



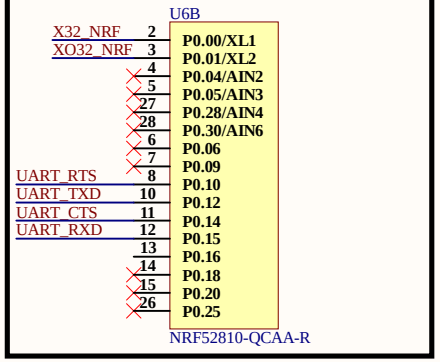
ANTENNA CONN



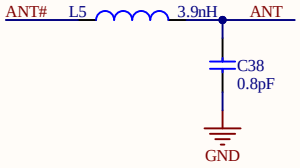
CRYSTALS

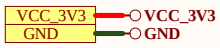


BLE TRANCIEVER I/O BANK



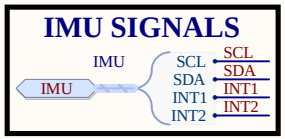
ANTENNA SUPPORTING CIRCUIT

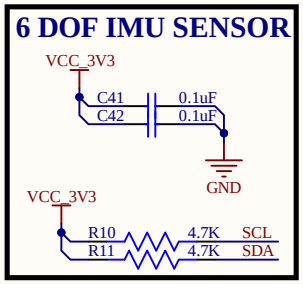


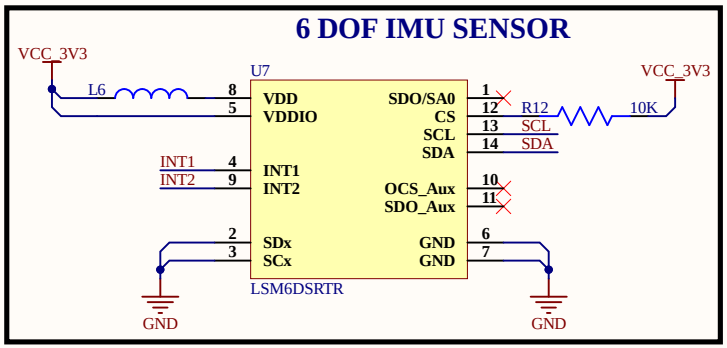


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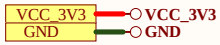
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Size	Number		Revision
A4	1		1
Date:	6/30/2022	Sheet 6 of	8
File:	C:\Users\..\BLE-Section.SchDoc	Drawn By:	Huzaifa Ali



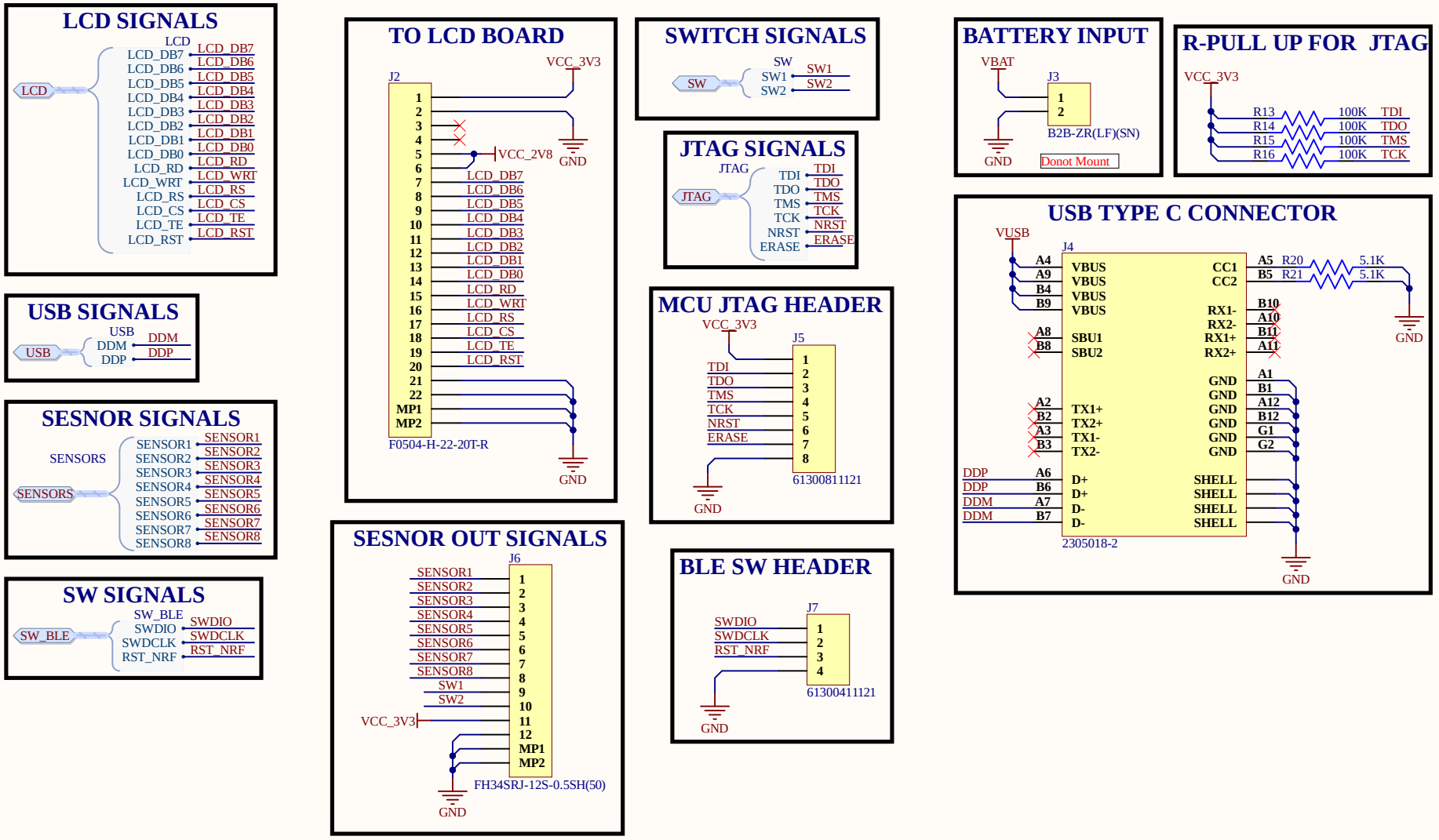




CS logic 1 to enable I2C and disable SPI mode.



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	Size: A4	Number: 1	Revision: 1
	Date: 6/30/2022	Sheet 7 of 8	
	File: C:\Users\..\IMU-Section.SchDoc	Drawn By: Huzaifa Ali	


LCD SIGNALS
LCD
LCD_DB7
LCD_DB6
LCD_DB5
LCD_DB4
LCD_DB3
LCD_DB2
LCD_DB1
LCD_DB0
LCD_RD
LCD_WRT
LCD_RS
LCD_CS
LCD_TE
LCD_RST
USB SIGNALS
USB
DDM
DDP
SESNOR SIGNALS
SENSORS
SENSOR1
SENSOR2
SENSOR3
SENSOR4
SENSOR5
SENSOR6
SENSOR7
SENSOR8
SW SIGNALS
SW_BLE
SWDIO
SWDCLK
RST_NRF
TO LCD BOARD
J2
VCC_3V3
VCC_2V8
GND
F0504-H-22-20T-R
SESNOR OUT SIGNALS
J6
VCC_3V3
GND
FH34SRJ-12S-0.5SH(50)
SWITCH SIGNALS
SW
SW1
SW2
JTAG SIGNALS
JTAG
TDI
TDO
TMS
TCK
NRST
ERASE
MCU JTAG HEADER
VCC_3V3
J5
GND
61300811121
BLE SW HEADER
J7
GND
61300411121
BATTERY INPUT
VBAT
J3
GND
B2B-ZR(LF)(SN)
Donot Mount
R-PULL UP FOR JTAG
VCC_3V3
R13
R14
R15
R16
100K
TDI
TDO
TMS
TCK
USB TYPE C CONNECTOR
VUSB
J4
VBUS
CC1
CC2
SBU1
SBU2
RX1-
RX2-
RX1+
RX2+
TX1+
TX2+
TX1-
TX2-
D+
D-
SHELL
GND
R20
R21
5.1K
DDP
DDM
2305018-2
VCC_2V8
VUSB
VCC_3V3
GND
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Title
ArmBand-Control-Board.PrjPcb
Size
A4
Number
1
Revision
1
Date:
6/30/2022
Sheet 8 of 8
File:
C:\Users\..\Connector-Section.SchDoc
Drawn By:
Huzaifa Ali